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ARTICLE

Competitive Position, Expectations, and Managerial Approaches to Risk in National Football League Football

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This paper analyzes how National Football League coaches incorporate information on competitive position and expectations of winning into fourth down decisions. Theoretically, teams should take more risk when their win probability is relatively low since additional risk broadens the distribution of possible outcomes thereby increasing the likelihood of victory. Our findings confirm this. We find that heavier favorites and teams having had higher levels of win probability within the game attempt more fourth down conversions, both on plays where attempting a play is estimated to be optimal and on plays where it is not. One possible explanation for this is coaches in those positions more likely frame their current win probability as being below a previously defined reference point, which reduces their degree of loss aversion. On balance, we find that this increased aggressiveness on fourth down improves the expected changes in win probability in this context.

Keywords: decision making, management effectiveness, manager, sports economics

When individuals or organizations face risky decisions, they consider factors, such as the likelihood of success, the gains associated with a successful outcome, the losses associated with an unsuccessful outcome, and any asymmetries between a relative gain and a relative loss. When success is defined by a rank order finish among competitors, the relative position of the competitor is another factor that must be considered. In such a setting, if a decision-maker is in a weak position relative to its competition, it may have an incentive to tolerate higher levels of risk—increasing the variance of the possible outcomes can increase the likelihood that the decision-maker finishes ahead of its rival. Furthermore, prior expectations about success, and changes in expectations can also factor into these decisions often in ways that comport with established behavioral phenomena. This paper empirically analyzes the degree to which organizations adjust their risk-taking behavior in the face of changing competitive positioning (relative to their rivals) and factors that might frame their expectations of winning by focusing on the fourth down decisions of teams in the National Football League (NFL). We then evaluate the degree to which sensitivity to these factors affects the efficiency of these decisions in this context.

This paper builds on theoretical literature from several streams. The first of these involves studies of attitudes toward risk in contests where rank order finish matters. The second has to do with behavioral phenomena that can impact risk-taking behavior. The third involves the particular inefficiencies associated with fourth down decisions in football. These factors each have direct implications of their own, but the interactions between them are of particular importance in this study.

In a tournament setting, there is a discrete marginal benefit associated with a competitor improving their rank order finish.[1] Because the rank order of finish is what matters in a head-to-head football game (finishing ahead of the opposing team is the goal, and there is a large, discrete increase in the team's benefit when it goes from just behind its opponent to just ahead of that opponent), it is applicable in the setting considered here. Rosen (1988) outlines numerous contexts where the rank order or relative position might matter in theory. Theoretically, agents should be willing to take on more risk when their relative position is weaker because the range of outcomes from the safer strategy has narrower tails. Cabral (2003) presents a model of dynamic competition in the context of choosing to spend research and development funds, and the equilibrium is consistent with this intuition: agents choose a risky project when behind in the race or a safe project when ahead of the competition. There is a body of empirical work that supports this theoretical work, which is discussed in "Background and Literature Review" section.

In terms of behavioral considerations, the well-known features of Kahneman and Tversky's (1979) prospect theory also inform some likely responses. For example, evaluating from a reference point, decision-makers are loss averse, their value function is asymmetric, and losses are felt more strongly than gains. If a decision-maker is below their reference point and in the "loss" frame, the calculus changes compared to a decision-maker at their reference point. The shape of the curvature of the value function in the loss frame implies that the utility benefits of a positive outcome are enhanced, and the downside of a negative outcome is blunted. Such a situation incentivizes more risk-taking.

In our context, it is also important to recognize that risk-taking can be influenced by prior experiences. For example, Thaler and Johnson (1990) find evidence of the "house money effect," whereby "after a gain, subsequent losses that are smaller than the original gain can be integrated with the prior gain mitigating the influence of loss aversion," and the "break even effect," whereby decisions which are able to overcome losses and "break even" (and thus allow a decision-maker to return to an initial reference point following a loss) are frequently chosen. In most settings, reference points and expectations are entirely unobserved so evaluating

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behavioral tendencies arising from reference points is not possible; sports settings, however, can be the exception to this.

Specifically within NFL football, it has been shown that coaches generally attempt too few plays on fourth downs if the goal is win maximization (Romer, 2006).[2] Romer (2006) describes this as an inefficiency, whereas subsequent research (e.g., Goff & Locke, 2019) provides evidence that risk aversion in a broader context than win maximization is also a factor. However, even if some decisions are justified by risk preferences, inefficiencies could certainly be the result of a challenging decision-making environment. In the NFL, teams must run a play in less than 40 s from the end of the previous play, or they receive a penalty. Coaches prepare for these decisions ahead of time, but they may take mental shortcuts that may be suboptimal. People often make different reflexive decisions than they would if they were able to slowly process all of the information in a rational way. Loewenstein (1996) suggests that the presence of visceral factors can generate seemingly inconsistent behavior in many contexts. Metcalfe and Jacobs (1998) discuss the impact of stress which leads people to rely on hot memory systems which produce reflexive decisions rather than cold systems that rely on logic. These factors are possible reasons why this inefficiency persisted throughout the sample period, as is discussed in more detail in the following section.

This paper addresses the factors related to fourth down risk-taking in the NFL and the relative efficiency of those decisions using play-by-play data. First, we estimate models to determine the optimal fourth down decision as a function of various situational factors including pregame expectations and a team's current likelihood of winning (as measured by Sports Reference's win probability model, which calculates, at any point within the game, a team's likelihood of winning based on observable characteristics of the game at that point in time). These models allow us to divide the data into fourth down plays where attempting a play is optimal and plays when it is not. For both sets of plays, we find that teams are sensitive to their competitive position, with teams in worse positions relative to their opponent (i.e., teams with a lower win probability) being more likely to attempt a fourth down play rather than punt or attempt a field goal. We then consider how expectations evolve during the game and determine whether prior expectations of victory earlier in the game have a significant influence on current in-game decisions. We observe that heavier pregame favorites and teams that have experienced higher expectations for victory earlier in the game are significantly more aggressive in fourth down in situations. This increased aggressiveness is especially pronounced for teams that have fallen from a very high win probability to a low one. This is consistent with the behavioral tendency described in the loss frame in prospect theory where decision-makers operating at a level below their reference point are more willing to take risks to try to recover losses. Consistent with the literature, we find that teams exhibit an inefficiency where too few fourth downs are attempted from the perspective of maximizing the average change in win probability on a given play. As such, anything that encourages marginally more aggressive risk-taking generally improves a team's likelihood of winning a given game.

The rest of this paper is organized as follows: "Background and Literature Review" section gives background on the related literature and the concept of win probability, "Data and Models" section describes the data and models, "Empirical Results and Discussion" section presents and discusses the empirical results, and "Conclusion" section concludes.

Background and Literature Review

Empirical Studies on Factors Affecting Risk Tolerance

The empirical work on risk-taking in competitive settings where rank order finish determines payoffs confirms the intuition of the theoretical work within this space, suggesting decision-makers are sensitive to variation in competitive conditions. Much of the literature shows how the level of variation in the convexity of the payoff structure affects risk-taking, with greater convexity leading to more risk-taking: Becker and Huselid (1992) show this effect in auto racing, while Kini and Williams (2012) and Coles et al. (2018) document this phenomenon in the competition for CEO positions. More directly relevant to this study, though, are the studies that specifically focus on how decision-makers respond to variation in their own competitive position. These studies show an increase in risk-taking when one's competitive position is poor due to the relatively large benefit associated with generating an outcome that improves one's rank order finish.

Along these lines, Ozbeklik and Smith (2017) find that in match-play golf, individuals take more risks when they are further behind and when they are of lower ability than their competitors. Similarly, Hood (2008) finds that golfers increase their level of risk when they are in an unfavorable competitive position and reduce their level of risk when they have a clear lead. Risky personnel decisions are also sensitive to competitive position. Barden and Choi (2021) show that Major League Baseball teams take greater risks by drafting high school players when the team is at a relative resource disadvantage, or when they are more successful. There are also empirical findings in settings outside of sports—in particular, within the mutual fund management industry—that are consistent with this pattern. Brown et al. (1996) show that funds with returns below the median level in the middle of the year shift to allocations that are more volatile than those that had returns above the median level at mid-year. Chevalier and Ellison (1997) also find that mutual funds managers adjust the riskiness of their holdings depending on their year-to-date return in response to nonlinearities in the payoffs associated with fund performance.[3]

Sports settings provide some examples where reference points can be empirically examined. Pope and Schweitzer (2011) find evidence of loss aversion in professional golfers' putting relative to a reference point of making a par. Coates et al. (2014) find that consumer choices related to game attendance is consistent with loss aversion. In the context of NFL football, we can observe important frames of reference, namely the pregame betting line (i.e., the point spread) and the in-game win probability. The variation in these quantifiable factors within NFL games provides another context in which this set of issues can be examined.

Deviations From the Status Quo in Sports

There are also a number of papers suggestive of mechanisms through which inefficiencies persist over time. This matters in this context since we use play-by-play data from NFL games from 2008 to 2018, a period just prior to a more widespread change in attitude toward fourth down decisions. The conventional wisdom that held for decades was that going for it was risky and most coaches were reluctant to attempt a conversion in favor of a kick. However, statistical analysis of fourth down decisions emerged with Romer (2006), who showed teams were attempting inefficiently few fourth

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down plays, and slowly began to gain traction as innovative coaches changed the way they approached the decision.

There is evidence in the literature that organizations experience agency problems that arise from gaps between understanding optimal decisions, implementing changes, and understanding the implications for personnel decisions. In the context of sports, Foreman et al. (2021) demonstrate that when the NFL changed the rules by penalizing actions that were deemed dangerous to players, coaches who do not adapt were penalized more often. However, the teams being penalized more often tended to win more games which conflicts with the goal to improve player safety. Yet, the coaches committing penalties experienced a lower probability of future employment as a head coach.

Spilker and Deutscher (2024) observed a strategic benefit for NBA teams from choosing the side of the court where the team will play offense in front of their own bench in the second half. Traditionally, teams choose to play the first half on offense in front of their own bench. Interestingly, in 2015–2016 22% of teams chose the optimal strategy and this increased to 33% for the 2022–2023 season. It seems that as teams start to understand the benefits of going against convention, they slowly adapt their decision making.

Betting Lines and Win Probability

This analysis considers the competitive conditions within a game which might influence a coach's reference point. One variable is the pregame betting line, which indicates the number of points by which one team is favored and is thus a clear indication of the relative strength of the teams prior to the start of the game. Betting markets are very large and competitive, so the betting line can be thought of as an indication of the aggregate beliefs of the relative team quality prior to the start of the game. The beliefs about which team will win change as the game progresses. We capture the game dynamics by focusing on Sports Reference's win probability model. This model calculates, at any point in the game, the likelihood of a team winning the game based on the pregame betting line and competitive conditions.[4] It essentially assumes that, in competitively neutral conditions, the final scoring margin of a game follows a normal distribution centered at the pregame betting line and with a *SD* based on estimates from years of prior games.

As the game progresses, the calculation is modified based on the actual score of the game, down, distance, and field position of the offensive team, and the time remaining (the distribution becomes tighter as the end of the game approaches), so this is a measure that is constantly changing throughout the game based on the outcome of each play. It has a clear interpretation in terms of the relative strength of a team's competitive position at a given point in time, and the change in this variable provides a direct means of assessing the effectiveness of a particular play in terms of its effect on the team's ultimate goal of winning the game. It also allows us to capture how a team's likelihood of winning compares to earlier points in the game and consider whether path dependence and changes relative to a reference point influence the actions taken.

In this section, we have outlined several factors and behavioral mechanisms which may influence fourth down decisions and potentially lead coaches to make suboptimal choices. Initially there may have been a gap in understanding optimal decisions due to the probabilistic nature of outcomes necessary to weigh costs and benefits. However, inefficiencies may persist even after optimal decisions are understood. Coaches may be reluctant to change, but behavioral factors including risk tolerance in a tournament setting, loss aversion, and perceived reference points for success can continue to influence decisions. We empirically explore the role of these factors using win probability and pregame betting lines as the basis for defining a team's competitive position and whether they are likely to view that position as being below their reference point.

Data and Models

This analysis uses play-level data from the 2008 through 2018 NFL seasons from sports-reference.com (data on win probability was unavailable for subsequent years). While it would be preferable to have data on these subsequent years, given this study's focus on how competitive and behavioral factors interact with baseline levels of risk-taking, the analysis remains instructive in that light even though four down attempt rates have increased from 2018 onward. The data includes information on the game situation (down, distance to go for a first down, field position, quarter of the game, time remaining in quarter), the betting line on the game (a negative number corresponds to a team being favored by that number of points, while a positive number corresponds to the number of points by which the opposing team is favored); the nature of the play (whether the team chose a running play, passing play, punt, or field goal attempt); the result of the play; and the postplay win probability value. Throughout this analysis, the variable *win probability* refers to the offense's win probability before the fourth down play, and the change in *win probability* is the win probability after the fourth down play minus the value before the play. This win probability data is used to construct a within-game measure of the offensive team's maximum level of win probability up to that point in the game. We also include data on the identity of the head coach and that coach's years of prior experience. This data also comes from sports-reference.com.

There are 37,692 total regular season fourth downs in the data set (excluding the final plays of either half). Though we examine several subsets of fourth down plays, we focus on the set of fourth downs where teams could expect a greater change in win probability from attempting a play on fourth down than from their other options. In order to do this, we run three models designed to predict, based on the game situation, the expected change in win probability from attempting a play, punting, or attempting a field goal, respectively. These models take the following form:

$$\begin{aligned}&\text{Change in win probability}_{it}\\&= \beta_0 + \beta_1(\text{win probability}_{it}) + \beta_2(\text{betting line}_{it})\\&+ \beta_3(\text{home team}_{it}) + \beta_4(\text{neutral site}_{it}) + \beta_5(\text{division rival}_{it})\\&+ \boldsymbol{\beta_6}(\textbf{field position}_{\mathbf{it}}) \times (\textbf{yards to go}_{\mathbf{it}}) + \boldsymbol{\beta_7}(\textbf{game quarter}_{\mathbf{it}})\\&+ \boldsymbol{\beta_8}(\textbf{lead}_{\mathbf{it}}) + \boldsymbol{\beta_9}(\textbf{game week}_{\mathbf{it}}) + \boldsymbol{\delta}_{\mathbf{t}} + \boldsymbol{\varepsilon}_{\mathbf{it}}.\end{aligned} \tag{1}$$

Variables are indexed by play (i) and season (t). *Home team*, *neutral site*, and *divisional rival* are dummy variables which indicate whether the possessing team is playing in its home stadium, whether the game is at a neutral site,[5] and whether the teams compete in the same four-team division, respectively. **Field position** is a series of indicator variables that describe distance from the end zone in 10-yard increments (i.e., 0–9 yards, 10–19 yards, etc.). **Yards to go** is a series of indicator variables for yards needed to earn a first down (one indicator for each number from one to nine, and a 10th dummy variable for situations where 10 or more yards are needed). A vector of the interactions of these dummy

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variables is created to provide for a more flexible set of controls than simply including the sets of indicators on their own (the uninteracted indicator variables are not included in the specification since these are collinear with the vector of interaction terms). The interaction of those two variables is included as a control. **Game quarter** is a series of four indicator variables representing the quarter of the game in which a decision is being made. **Lead** is the possessing team's lead or deficit at the time of the decision, represented by a series of indicator variables (trailing by 9 or more points, trailing by between 4 and 8 points, trailing by between 1 and 3 points, game tied, leading by between 1 and 3 points, leading by between 4 and 8 points, leading by 9 or more points). **Game week** is a series of indicator variables representing each of the 17 weeks in an NFL season. The term δ_t is a vector of season-specific fixed effects.

These models are estimated using observed plays when a particular decision was made and thus a particular outcome was realized. Based on these estimates, we can determine whether a fourth down attempt is on average the choice that maximizes win probability for a given set of situational factors. This allows us to consider several subsets of the data—in some cases, we consider all fourth downs, but we also consider the subset of fourth down plays for which the estimates of the models indicate that attempting a play would be optimal in terms of win probability and the subset of fourth down plays for which attempting a play would be not be optimal. The subset of plays for where the models indicate that attempting a play on fourth down would improve the expected win probability relative to punting or attempting a field goal includes 18,366 fourth down observations (though in the regression models, this drops to 18,108 due to missing data on certain covariates).

As robustness checks, in some specifications, we make additional adjustments to the subset of the data. In some specifications, we eliminate plays with extreme win probability values, which are defined as plays where the offensive team has a win probability below 10% or above 90%, to make sure that the findings are not being driven by activity on the extreme ends of this distribution.[6] In other specifications, we restrict the analysis to the 2013–2018 seasons. This is because pro-football-reference began publishing these win probability values beginning in the 2013 season (Paine, 2013). It should be noted that other media outlets were applying this concept to NFL beginning in 2008, and this basic concept had been in use in other sports, in particular baseball, for many years before this (Burke, 2008). As a result, the paradigm for fourth down decision making was understood and quantified to some extent for all the years in the data set. Finally, we run some specifications where the expected change in win probability from the optimal choice is more than a full percentage point higher than what could be expected from the next best alternative. This removes observations where the optimal choice is less clear-cut statistically.

We use a variety of probit specifications to estimate the effects that the level of win probability and factors that would raise expectations of winning have on fourth down choices. The basic model is:

$$\begin{aligned}&\text{Fourth down attempt}_{\text{ict}}\\&= \beta_0 + \beta_1(\text{win probability}_{\text{ict}}) + \beta_2(\text{betting line}_{\text{ict}})\\&+ \beta_3(\text{max win probability}_{\text{ict}}) + \beta_4(\text{neutral site}_{\text{it}})\\&+ \beta_5(\text{division rival}_{\text{it}}) + \boldsymbol{\beta_6}(\textbf{home team}_{\textbf{it}})\\&+ \boldsymbol{\beta_7}(\textbf{field position}_{\textbf{it}}) \times (\textbf{yards to go}_{\textbf{it}}) + \boldsymbol{\beta_8}(\textbf{game quarter}_{\textbf{it}})\\&+ \boldsymbol{\beta_9}(\textbf{lead}_{\text{it}}) + \boldsymbol{\beta_{10}}(\textbf{game week}_{\textbf{it}}) + \boldsymbol{\beta_{11}}(\textbf{coach prior exp}_{\textbf{ct}})\\&+ \boldsymbol{\beta_{12}}(\textbf{coach prior exp squared}_{\textbf{ct}}) + \boldsymbol{\delta}_{\ \textbf{t}} + \boldsymbol{\gamma_c} + \boldsymbol{\varepsilon_{it}}.\end{aligned} \tag{2}$$

The dependent variable is a binary variable equaling one if the team attempts a fourth down conversion and zero otherwise. Many of the control variables are the same as the ones in Equation (1), but there are a few differences. There are no controls related to individual coaches in Equation (1), but here there are (indeed, the data here is also indexed by [c for the individual coach in addition to being index by play [i] and season [t]). In particular, there are coach-specific fixed effects to control for the variation between coaches in their baseline level of fourth down aggressiveness (γ_c) as well as controls for a coach's years of head coaching experience and the square that value (as is done is Goodall et al., 2011 and Roach and Owens, 2024). In these models, we also create a vector of interaction terms between individual teams and whether they are the home team in order to account for differences in home field advantage across teams (studies such as Benz et al's., 2024 describe sources of variation in home field advantage). The other variable, which is a focus of the analysis is *max win probability*, which is the highest win probability level the team has experienced up until the particular fourth down being observed. Theoretically, conditional on the factors, such as the actual level of win probability, the betting line, and game situation, this variable should not affect the relative costs and benefits of attempting a play on fourth down (in terms of win probability), but it can plausibly affect these decisions through behavioral channels.

Empirical Results and Discussion

Figure 1 shows that in the primary subset of plays used in the analysis, teams attempt to run plays between 10% and 15% of the time. These relatively low percentages confirm earlier findings in this literature, such as Romer (2006), and suggest that this is a sufficiently wide set of fourth down decisions to conduct much of the analysis.

Table 1 describes the primary subset of fourth down plays. Within this set of plays, teams attempt a fourth down conversion 12.2% of the time, converting on 64.6% of those attempts. The average win probability for a coach facing a fourth down decision is 41.1%. On average, teams have experienced an in-game maximum win probability of 64.2%. The change in win probability on fourth down is close to zero, but the *SD* of 0.056 shows that the outcome of these fourth down plays has a material effect on the outcome of the game. The table also shows that coaches average 5.15 years of prior NFL head coaching experience.

The key relationships in the analysis are those between fourth down activity and the competitive state of the game, as measured by win probability itself, as well as other factors that might affect a team's expectation of winning. These factors include the betting line and the maximum prior level of in-game win probability, and we assume these are correlated with a team's reference point for success. All else equal, we expect teams that are in a more desperate competitive position to take more risks thereby broadening the range of possible outcomes. Furthermore, we expect teams that, all else equal (including the current level of win probability), have a greater expectation of winning to be more confident in their prospects of success on a given play and more willing to take risks when they are clearly performing below their expected level (that is, when they are in the loss frame as understood in prospect theory). Teams in these situations are thus expected to be more aggressive on fourth down.

Table 2 shows clear descriptive evidence that the current win probability factors into the fourth down decision and the associated changes in win probability. Teams in the lowest win probability bin

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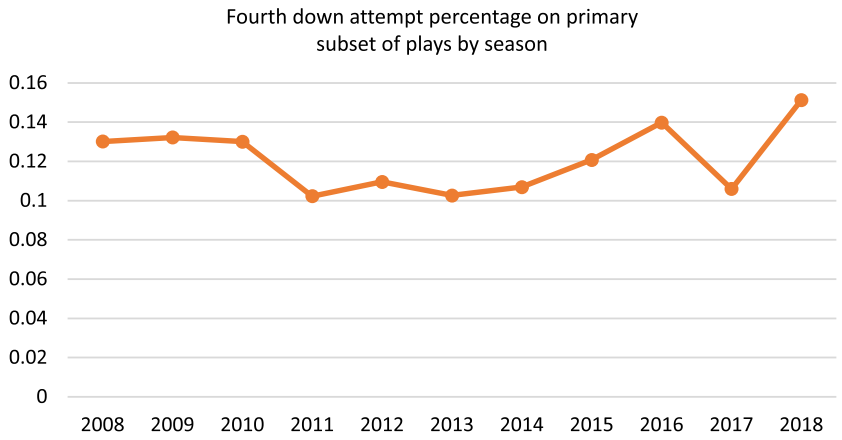

Figure 1 — Fourth down attempt percentages by season. *Source*: sports-reference.com. Statistics from the primary subset of fourth down plays in the data (only includes situations where attempting a play is associated with a greater expected improvement in win probability than a punt or field goal attempt); excludes the last play of the first half, and the last play of the game.

Table 1 Summary Statistics on Primary Subset of Fourth Down Plays

Variable	Obs	Mean	*SD*	Min	Max
Fourth down attempt	18,366	0.1218012	0.3270649	0	1
Fourth down conversion	2,237	0.6459544	0.47833	0	1
Win probability	18,366	0.4112378	0.2902934	0.001	0.999
Win probability 0%–25%	18,366	0.353207	0.4779793	0	1
Win probability 25%–50%	18,366	0.2790482	0.4485435	0	1
Win probability 50%–75%	18,366	0.2095176	0.4069754	0	1
Win probability 75%–100%	18,366	0.1582272	0.3649638	0	1
Win probability between 10% and 90%	18,366	0.7473593	0.4345385	0	1
Max win probability	18,366	0.6420901	0.2225562	0.024	0.999
Max win probability ≤25%	18,366	0.0379506	0.1910819	0	1
Max win probability between 25% and 50%	18,366	0.2602635	0.4387903	0	1
Max win probability between 50% and 75%	18,366	0.3522269	0.4776772	0	1
Max win probability > 75%	18,366	0.349559	0.4768437	0	1
Change in win probability	18,366	0.0120932	0.0557492	–0.888	0.907
Betting line	18,366	–2.196123	5.895614	–26.5	15.5
Betting line favorite	18,366	0.4243167	0.4942523	0	1
home team	18,366	0.5002722	0.5000135	0	1
Away team	18,366	0.4961886	0.4999991	0	1
Neutral site	18,366	0.0035391	0.059387	0	1
Divisional rival	18,366	0.345802	0.475642	0	1
Yards to go for first down	18,366	7.297343	5.838274	1	48
Yards to end zone	18,366	56.31134	21.83167	1	99
Lead	18,366	–1.814222	8.038656	–44	38
Pass	18,366	0.0645214	0.2456861	0	1
Rush	18,366	0.0572798	0.2323827	0	1
Field goal attempt	18,366	0.1443973	0.3515017	0	1
Punt	18,366	0.7327126	0.4425556	0	1
First quarter	18,366	0.2870522	0.4523985	0	1
Second quarter	18,366	0.2734945	0.4457646	0	1
Third quarter	18,366	0.1991724	0.3993888	0	1
Fourth quarter	18,366	0.2311336	0.4215691	0	1
Coach cumulative experience	18,366	5.147392	5.290175	0	23

Note. Source: sports-reference.com. Statistics from the primary subset of fourth down plays in the data (only includes situations where attempting a play is associated with a greater expected improvement in win probability than a punt or field goal attempt); excludes the last play of the first half, and the last play of the game.

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Table 2 Fourth Down Decisions and Associated Average Changes in Win Probability, By Win Probability Category

Offensive team's win probability	*N*	Likelihood of attempting fourth down play	Success rate when a play is attempted	Average change in win probability on attempt	Average change in win probability on nonattempts	Difference
Win probability 0%– 25%	6,487	0.182	0.631	0.030	0.009	0.030***
Win probability 25%– 50%	5,125	0.093	0.654	0.041	0.015	0.041***
Win probability 50%–75%	3,848	0.089	0.683	0.037	0.011	0.037***
Win probability 75%–100%	2,906	0.082	0.649	0.014	–0.001	0.014***
All categories	18,366	0.122	0.646	0.032	0.009	0.032***

Note. Source: sports-reference.com. Statistics from the primary subset of fourth down plays in the data (only includes situations where attempting a play is associated with a greater expected improvement in win probability than a punt or field goal attempt); excludes the last play of the first half, and the last play of the game.
***$p < .01$.

(those with a win probability between 0% and 25%) take the risk of attempting a fourth down conversion 18.2% of the time while teams in the highest win probability bin (those with a win probability between 75% and 100%) attempt a conversion 8.2% of the time. As one might expect, changes in win probability on a play depend in part on its starting level. In this subset of plays, teams at all levels of win probability significantly increase their expected win probability on average by attempting a play (this average includes both successful and unsuccessful attempts), the magnitude of the effect is lower for the highest levels of win probability (those levels above 75%). This is likely due to the fact that as a team's win probability rises, there is less room to improve it.

While Table 2 suggests a baseline level of inefficiency in fourth down decision making within the primary subset of fourth downs considered, these decisions might also be influenced by a team's expectation of success, even controlling for the current competitive state of the game. The likelihood of a fourth down attempt may also depend on the team's pregame expectations of victory as well as prior expectations within the game. Table 3 describes the likelihood of attempting a fourth down for various levels of win probability depending on whether the team is the pregame favorite, or not, based on the betting line. Table 4 displays the likelihood of an attempt depending on the highest level of win probability that the team has experienced at a prior point in the game. The results suggest that teams with higher expectations of winning attempt fourth down conversions at a higher rate. Even though pregame favorites tend to be making fourth down decisions from relatively high levels of win probability (which would reduce the likelihood of an attempt, all else equal), they attempt a play on fourth down at higher rates than underdogs. Conditioning on win probability level, we see that this difference is especially pronounced when teams face such a decision at a point where their win probability is low. When a team's win probability is between 0% and 25%, the favorites attempt a conversion 27.1% of the time, while the underdog attempts a conversion 15.8% of the time. This difference is significant at the 1% level. When looking at a team's maximum level of win probability, the pattern of increased fourth down aggressiveness, especially at low levels of win probability, persists. For teams facing a win probability between 0% and 25%, the ones who have experienced a win probability of 75% or higher at some prior point in the game attempt a conversion 35.1% of the time, a rate almost four times greater than the 9.6% for teams whose maximum win probability never exceeded 25%. In light of the results of Table 2, this increased aggressiveness in these fourth down situations likely increases a team's chances of ultimately winning the game.

Table 5 presents marginal effects for several specifications of the model presented in Equation (2), including some where the *max win probability* variable is omitted. This table provides a summary of the key effects that are consistent across various subsets in the data. In particular, with all else equal, teams attempt fourth down plays at significantly higher rates when their win probability falls, teams that are more heavily favored to win attempt fourth down plays at significantly higher rates, and teams that have experience higher levels of win probability earlier in the game attempt fourth down plays at significantly higher rates. This is true when looking at all situations (Specifications 1 and 2), situations where attempting a play is optimal based on our estimates (Specifications 3 and 4), and situations where attempting a play is not optimal according to our estimates (Specifications 5 and 6). The sensitivity to these factors increases when considering just the set of plays where running a play is the optimal choice based on the expected change in win probability. A 10 percentage point increase in *win probability* reduces the likelihood of attempting a fourth down play by 2.67 percentage points when looking the set of plays where a fourth down play is optimal (Specification 4), compared with a 1.31 percentage point reduction within the set of plays where a fourth down play is not optimal (Specification 6). A team being favored by an additional point increases the likelihood of a fourth down attempt by 0.262 percentage points in the set of plays where running a play is optimal and by 0.081 percentage points otherwise. A 10 percentage point increase in *max win probability* increases the likelihood of attempting a fourth down play by 0.539 percentage points in the set of plays where running a play is optimal and by 0.426 percentage points otherwise.

Table 6 reports marginal effects for specifications where our estimates suggest attempting a fourth down play will have on average a more positive impact on win probability than punting or attempting a field goal. Each specification, regardless of any adjustment to the criteria for including particular observations, shows a clear and significant negative relationship between *win probability* and the likelihood of attempting a fourth down attempt (in each specification, the effect is significant at the 1% level). In Specification 1, the estimated coefficient on *win probability*

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Table 3 Differences in Likelihood of Fourth Down Attempt, By Win Probability Category, Comparing Betting Line Favorites and Nonfavorites

Offensive team's win probability	Pregame favorite		Not pregame favorite		Difference in likelihood of attempt
	N	Likelihood of fourth down attempt	*N*	Likelihood of fourth down attempt	
Win probability 0%–25%	1,361	0.271	5,126	0.158	0.113***
Win probability 25%–50%	1,694	0.116	3,431	0.082	0.034***
Win probability 50%–75%	2,602	0.088	1,246	0.089	–0.001
Win probability 75%–100%	2,136	0.090	770	0.061	0.029**
All categories	7,793	0.127	10,573	0.118	0.008*

Source: sports-reference.com. Statistics from the primary subset of fourth down plays in the data (only includes situations where attempting a play is associated with a greater expected improvement in win probability than a punt or field goal attempt); excludes the last play of the first half, and the last play of the game.
*$p < .1$. **$p < .05$. ***$p < .01$.

suggests an increase of 10 percentage points in win probability reduces the likelihood of attempting a fourth down by 2.67 percentage points. This is consistent with teams in a more precarious competitive positions being more willing to take risks.

The *betting line* variable is also significant at the 1% level in each specification. In terms of magnitude, according to Specification 1, being favored by an additional point according to the betting line corresponds to a roughly 0.26 percentage point increase in a team's willingness to attempt a fourth down conversion, conditional on other factors. This effect could be due to relatively high expectations of a successful conversion, a relatively low degree of loss aversion, or both. Since the betting line generally reflects an underlying quality differential between the teams, it seems natural that it would affect a team's beliefs regarding the likelihood of a successful conversion. Also, conditional on the level of win probability, a team being more heavily favored means the team is more likely to be below their reference point of expected success (or further below that reference point on average), and being in that position makes them less loss averse.

The *max win probability* variable is also significant in each specification. The point estimate in Specification 1 suggests an increase of 10 percentage points in the maximum win probability increases the likelihood of attempting a fourth down by 0.54 percentage points. Since these estimated effects are conditional on the pregame betting line, if they arise from relatively high expectations about the likelihood of a successful conversion, those raised expectations come from factors other than the perceived pregame quality differential between the teams. Alternatively, to the extent that success early in the game raises a team's reference point for success, this effect could be due to reduced levels of loss aversion consistent with experiencing being in the loss frame in prospect theory. Taken together with the estimates related to the betting line, these estimates clearly indicate that expectations of success affect the decision in a manner consistent with prospect theory.

The state of the game (as measured by win probability) and the other factors affecting the fourth down decision may interact with one another. In particular, for relatively low levels of win probability, the factors related to a team's expectation of winning may become more pronounced (the sense of desperation might be more pronounced in such a situation). In order to investigate this, we interact *win probability* with the *betting line* and *max win probability*. The marginal effects from these models are presented in Table 7 (with robustness checks analogous to those in Table 6). Intuitively, it makes sense to think about how teams respond to a reduction in win probability since that is when a behavior-based reduction in loss aversion is more likely. The negative and significant coefficient on the interaction between *betting line* and *win probability* suggests that being a heavier favorite (a betting line where the negative number has a greater absolute value) means a team is less responsive to a reduction in win probability as it relates to attempting a fourth down play. This is perhaps in part due to the fact that heavier favorites are more aggressive on fourth downs at high levels of win probability (as was shown in Table 3). As it relates to *max win probability*, Specification 1 suggests that a higher *max win probability* still has a direct positive effect on a team's likelihood of attempting a fourth down, and the team is also more sensitive to changes in *win probability*. The interaction terms suggest for a team with a 10 percentage point higher value of *max win probability* would respond to a 10 percentage point reduction in win probability by being 0.2 percentage points more likely to attempt a fourth down play. This is consistent with the descriptive results in Table 4.

The results of Tables 6 and 7 describe specific situations where teams are more likely to attempt fourth down plays in a subset of plays for which that would generally improve their expected win probability. As such, this increased aggressiveness is a sign of increased efficiency. Increased aggressiveness on fourth down for plays where that is not the case (i.e., a punt or a field goal attempt increases expected win probability by more than attempting a play) could lead to reduced efficiency. Table 8 reports the marginal effects of the models analogous to those in Table 6, but for the subset of plays where our estimates suggest running a play is suboptimal. The results generally have the same direction as those in Table 6: lower *win probability*, being a heavier favorite, and having a higher *max win probability* are associated with attempting more fourth down plays, suggesting inefficiency in these decisions. The finding that increased aggressiveness on fourth down is associated with these factors in a variety of contexts does support the notion that these decision-makers are behaving in a way consistent with reduced loss aversion manifesting when their competitive position deteriorates relative to a reference point.

The point estimates in Table 8 are generally smaller than those reported in Table 6 (for instance a 10 percentage point increase in win probability is associated with a 1.3 percentage point increase in the likelihood of attempting a play on fourth down in the set of plays where it is not optimal to do so, compared with an increase of 2.7 percentage points in the set of plays where it is optimal to run a play), so there is reason to believe that, on balance, increased aggressiveness is efficient from the perspective of maximizing *win*

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Table 4 Differences in Likelihood of Fourth Down Attempt, By Win Probability Category and In-Game Maximum Win Probability

	Likelihood of fourth down attempt			
Offensive team's win probability	**Max WP ≤25%**	**25% < Max WP ≤50%**	**50% < Max WP ≤75%**	**Max WP > 75%**
Win probability 0%–25%	0.096	0.138	0.210	0.351
Win probability 25%–50%		0.084	0.094	0.110
Win probability 50%–75%			0.094	0.083
Win probability 75%–100%				0.082
All categories	0.096	0.117	0.131	0.119

Note. Source: sports-reference.com. Statistics from the primary subset of fourth down plays in the data (only includes situations where attempting a play is associated with a greater expected improvement in win probability than a punt or field goal attempt); excludes the last play of the first half, and the last play of the game. Max WP = maximum win probability.

Table 5 Marginal Effects From Probit Regressions of Attempting a Fourth Down Play on Win Probability Measures and Measures of Win Expectation, Various Subsets

Variables	**(1) Fourth down attempt**	**(2) Fourth down attempt**	**(3) Fourth down attempt**	**(4) Fourth down attempt**	**(5) Fourth down attempt**	**(6) Fourth down attempt**
Win probability	–0.186***	–0.195***	–0.259***	–0.267***	–0.121***	–0.131***
	(0.0119)	(0.0118)	(0.0185)	(0.0188)	(0.0139)	(0.0138)
Betting line	–0.00237***	–0.00150***	–0.00356***	–0.00262***	–0.00146***	–0.000809***
	(0.000230)	(0.000268)	(0.000377)	(0.000433)	(0.000253)	(0.000293)
Max win probability		0.0522***		0.0539***		0.0426***
		(0.00841)		(0.0132)		(0.0101)
All fourth downs	X	X				
Fourth downs where attempting a play increases win probability more than other options			X	X		
Fourth downs where attempting a play does not increase win probability more than other options					X	X
Observations	37,320	37,320	18,108	18,108	18,894	18,894

Note. Source: sports-reference.com. Excludes the last play of the first half and last play of the game. Robust *SE*s in parentheses.
***$p < .01$.

probability, but we attempt to quantify this with ordinary least squares regressions of the following basic form:

$$\begin{aligned}
&\text{Change in win probability}_{\text{ict}} \\
&= \beta_0 + \beta_1(\text{fourth down attempt}_{\text{ict}}) \\
&+ \beta_2(\text{fourth down attempt}_{\text{ict}}) \times (\text{win probability}_{\text{ict}}) \\
&+ \beta_3(\text{fourth down attempt}_{\text{ict}}) \times (\text{betting line}_{\text{ict}}) \\
&+ \beta_4(\text{fourth down attempt}_{\text{ict}}) \times (\text{max win probability}_{\text{ict}}) \\
&+ \beta_5(\text{win probability}_{\text{ict}}) + \beta_6(\text{betting line}_{\text{ict}}) \\
&+ \beta_3(\text{max win probability}_{\text{ict}}) \\
&+ \beta_4(\text{neutral site}_{\text{it}}) + \beta_5(\text{division rival}_{\text{it}}) + \beta_6(\text{home team}_{\text{it}}) \\
&+ \beta_7(\text{field position}_{\text{it}}) \times (\text{yards to go}_{\text{it}}) \\
&+ \beta_8(\text{game quarter}_{\text{it}}) + \beta_9(\text{lead}_{\text{it}}) \\
&+ \beta_{10}(\text{game week}_{\text{it}}) + \beta_{11}(\text{coach prior exp}_{\text{ct}}) \\
&+ \beta_{12}(\text{coach prior exp squared}_{\text{ct}}) + \delta_{\text{t}} + \gamma_c + \epsilon_{\text{it}}.
\end{aligned} \tag{3}$$

The dependent variable is the *change in win probability*, and the goal of the regression is to determine if increase fourth down attempts, on balance (i.e., for the set of all fourth down plays), lead to more positive changes in win probability compared with what would be expected had a different choice been made. The model essentially compares the expected change in win probability from a fourth down attempt (conditional on the covariates) to the expected change in win probability from a punt or attempted field goal, based on the relatively likelihood of those choices being made (conditional on the covariates). Interactions between *fourth down attempts* and other variables associated with changes in the likelihood of attempting a play (*win probability*, *betting line*, *max win probability*) are included to determine if there are any specific mechanisms by which efficiencies or inefficiencies in decision making present themselves. The other covariates are the same as in Equation (2). These models, presented in Table 9, are estimated for all the fourth down plays in the data regardless of whether a fourth down attempt is expected to increase *win probability* relative to other options.

The estimates from Table 9 show that, consistent with Romer (2006), teams attempt too few fourth downs relative to the level that would maximize win probability; there is a positive coefficient on *fourth down attempt* in each of the specifications, and in four of the six specifications, it is statistically significant. Specification 1 suggests that, conditional on the game situation, attempting a

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Table 6 Marginal Effects From Probit Regressions of Attempting a Fourth Down Play on Win Probability Measures and Measures of Win Expectation (When a Play Raises Expected Win Probability)

Variables	(1) Fourth down attempt	(2) Fourth down attempt	(3) Fourth down attempt	(4) Fourth down attempt	(5) Fourth down attempt	(6) Fourth down attempt
Win probability	–0.267***	–0.177***	–0.228***	–0.257***	–0.173***	–0.219***
	(0.0188)	(0.0197)	(0.0241)	(0.0209)	(0.0219)	(0.0272)
Betting line	–0.00262***	–0.00227***	–0.00225***	–0.00211***	–0.00197***	–0.00193***
	(0.000433)	(0.000458)	(0.000589)	(0.000484)	(0.000497)	(0.000671)
Max win probability	0.0539***	0.0549***	0.0414**	0.0640***	0.0583***	0.0375*
	(0.0132)	(0.0145)	(0.0179)	(0.0145)	(0.0155)	(0.0196)
Excludes WP < 0.1 and WP > 0.9		X			X	
Only includes seasons after 2012			X			X
Attempting play increases expected WP by 0.01 relative to next best option				X	X	X
Observations	18,108	13,009	9,329	13,753	9,944	7,036

Note. Source: sports-reference.com. Statistics from the primary subset of fourth down plays in the data (only includes situations where attempting a play is associated with a greater expected improvement in win probability than a punt or field goal attempt); excludes the last play of the first half, and the last play of that game. Robust *SE*s in parentheses.
$^{*}p < .1$. $^{**}p < .05$. $^{***}p < .01$.

Table 7 Marginal Effects From Probit Regressions of Attempting a Fourth Down Play on Win Probability Measures, Measures of Win Expectation, and Interactions (When a Play Raises Expected Win Probability)

Variables	(1) Fourth down attempt	(2) Fourth down attempt	(3) Fourth down attempt	(4) Fourth down attempt	(5) Fourth down attempt	(6) Fourth down attempt
Win probability	–0.106**	0.0456	–0.0661	–0.0443	0.0533	–0.00853
	(0.0472)	(0.0510)	(0.0619)	(0.0524)	(0.0544)	(0.0684)
Betting line	0.000415	0.000956	0.000890	0.000961	0.00134*	0.00110
	(0.000567)	(0.000787)	(0.000748)	(0.000610)	(0.000810)	(0.000801)
Max win probability	0.116***	0.139***	0.103***	0.133***	0.143***	0.104***
	(0.0164)	(0.0228)	(0.0219)	(0.0180)	(0.0236)	(0.0238)
Win probability × Betting line	–0.00763***	–0.00693***	–0.00809***	–0.00806***	–0.00733***	–0.00818***
	(0.00111)	(0.00155)	(0.00153)	(0.00126)	(0.00162)	(0.00178)
Win probability × Max win probability	–0.209***	–0.277***	–0.210***	–0.270***	–0.284***	–0.268***
	(0.0510)	(0.0607)	(0.0674)	(0.0565)	(0.0632)	(0.0748)
Excludes WP < 0.1 and WP > 0.9		X			X	
Only includes seasons after 2012			X			X
Attempting play increases expected WP by 0.01 relative to next best option				X	X	X
Observations	18,108	13,009	9,329	13,753	9,944	7,036

Note. Source: sports-reference.com. Statistics from the primary subset of fourth down plays in the data (only includes situations where attempting a play is associated with a greater expected improvement in win probability than a punt or field goal attempt); excludes the last play of the first half, and the last play of that game. Robust *SE*s in parentheses.
$^{*}p < .1$. $^{**}p < .05$. $^{***}p < .01$.

fourth down play increases the expected *change in win probability* by 1.56 percentage points relative to plays where there was a punt play or field goal attempt. This suggests that the predominant inefficiency in these decisions is attempting too few plays on fourth down, and additional fourth down attempts as well as characteristics associated with increased fourth down attempts (such as being favored more heavily and having a relatively high *max win probability*) tend to raise the expected *change in win probability* on balance, even if it is associated with taking some risks at time where another option would be optimal. The interactions between *fourth down attempt* and the key explanatory variables in our models of risk-taking suggest that there is no specific efficiency or

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Table 8 Marginal Effects From Probit Regressions of Attempting a Fourth Down Play on Win Probability Measures and Measures of Win Expectation (When a Play Lowers Expected Win Probability)

Variables	(1) Fourth down attempt	(2) Fourth down attempt	(3) Fourth down attempt	(4) Fourth down attempt	(5) Fourth down attempt	(6) Fourth down attempt
Win probability	–0.131***	–0.0741***	–0.147***	–0.0830***	–0.0517***	–0.102***
	(0.0138)	(0.0154)	(0.0187)	(0.0127)	(0.0153)	(0.0187)
Betting line	–0.000809***	–0.000632*	–0.000969**	–0.000387	–0.000249	–0.000570
	(0.000293)	(0.000344)	(0.000406)	(0.000270)	(0.000346)	(0.000403)
Max win probability	0.0426***	0.0462***	0.0422***	0.0305***	0.0353***	0.0378***
	(0.0101)	(0.0119)	(0.0129)	(0.00936)	(0.0121)	(0.0132)
Excludes WP < 0.1 and WP > 0.9		X			X	
Only includes seasons after 2012			X			X
Attempting play decreases expected WP by 0.01 relative to next best option				X	X	X
Observations	18,894	12,068	10,497	12,892	7,761	6,701

Note. Source: sports-reference.com. Statistics from the subset of fourth down plays in the data that excludes situations where attempting a play is associated with a greater expected improvement in win probability than a punt or field goal attempt); excludes the last play of the first half, and the last play of that game. Robust *SE*s in parentheses.
$^{*}p < .1$. $^{**}p < .05$. $^{***}p < .01$.

Table 9 Ordinary Least Squares Regression of Change in Win Probability on Fourth Down Attempts, Win Probability, Measures of Win Expectation (All Fourth Downs)

Variables	(1) Change in win probability	(2) Change in win probability	(3) Change in win probability	(4) Change in win probability	(5) Change in win probability	(6) Change in win probability
Fourth down attempt	0.0156*	0.0385***	0.0244**	0.0115	0.0337**	0.0199
	(0.00850)	(0.0135)	(0.0119)	(0.00987)	(0.0152)	(0.0138)
Fourth down attempt × Win probability	0.000711	–0.00354	0.00640	–0.00552	–0.0222	0.00307
	(0.00700)	(0.0155)	(0.00925)	(0.00822)	(0.0176)	(0.0108)
Fourth down attempt × Betting line	0.000257	–0.000535	–0.000418	0.000382	–0.000389	–0.000255
	(0.000431)	(0.000637)	(0.000638)	(0.000512)	(0.000744)	(0.000769)
Fourth down attempt × Max win probability	–0.00318	–0.0307	–0.0241	0.0126	–0.00284	–0.0103
	(0.0158)	(0.0256)	(0.0218)	(0.0181)	(0.0282)	(0.0248)
Win probability	–0.112***	–0.129***	–0.100***	–0.125***	–0.145***	–0.108***
	(0.00629)	(0.0106)	(0.00834)	(0.00746)	(0.0123)	(0.00954)
Betting line	–0.00127***	–0.00182***	–0.000990***	–0.00151***	–0.00213***	–0.00115***
	(0.000117)	(0.000212)	(0.000153)	(0.000142)	(0.000251)	(0.000184)
Max win probability	0.0185***	0.0220***	0.0212***	0.0201***	0.0256***	0.0235***
	(0.00366)	(0.00551)	(0.00524)	(0.00419)	(0.00629)	(0.00591)
Excludes WP < 0.1 and WP > 0.9		X			X	
Only includes seasons after 2012			X			X
Attempting play increases expected WP by 0.01 relative to next best option				X	X	X
Observations	37,564	26,923	20,455	27,286	19,593	14,692
R-squared	0.080	0.078	0.073	0.099	0.097	0.089

Note. Source: sports-reference.com. Statistics from the primary subset of fourth down plays in the data (only includes situations where attempting a play is associated with a greater expected improvement in win probability than a punt or field goal attempt); excludes the last play of the first half, and the last play of that game. Robust *SE*s in parentheses.
$^{*}p < .1$. $^{**}p < .05$. $^{***}p < .01$.

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inefficiency associated with sensitivity to those factors outside of the direct effect they have on the overall attitude toward these risky decisions.

Conclusions

This paper seeks to analyze how competitive positioning within a contest and factors related to expectations of success affect risk-taking behavior using data on NFL fourth down decisions. Studies such as Romer (2006) have shown an inefficiency in these types of decisions; teams generally attempt too few plays on fourth down relative to what would maximize their chances of winning a given game. This paper considers the theoretical work on rank order contests, where the literature suggests teams in precarious positions should invite more risk, and the literature on prospect theory, which suggests decision-makers are less loss averse when they are below their reference point, to empirically investigate whether the patterns of behavior comport with this literature. We then consider the implications for the efficiency of the decisions being made in this context.

Using NFL play-by-play data, we find that a team's competitive position, as summarized by its win probability at the point at which it must make a fourth down decision, significantly impacts these choices in the manner predicted by economic theory: teams take significantly more risks when their competitive position is relatively poor, controlling for the pregame betting expectations and other situational factors. Specifically, a decrease of 10 percentage points in win probability being associated with an increase of 2.67 percentage points in the likelihood of attempting a fourth down play for the subset of plays for which attempting a first down is the optimal choice and a 1.31 percentage point increase for the subset of plays for which attempting a fourth down play is not the optimal choice. We also find that heavier favorites and teams that have experienced higher win probabilities are significantly more likely to attempt fourth down plays. Being favored by an additional point according to the betting line corresponds to a 0.26 percentage point increase in a team's willingness to attempt a fourth down play when attempting a play is optimal and a 0.08 percentage point increase when a field goal attempt or punt is optimal. An increase in the maximum level of win probability a team has experienced within the game is also associated with attempting fourth down conversions at a higher rate; conditional on win probability and the betting line an increase of 10 percentage points in the maximum win probability increases the likelihood of attempting a fourth down by 0.54 percentage points when a fourth down play is the optimal decision and by 0.43 otherwise. The increased fourth down aggressiveness associated with a team's maximum win probability is most pronounced for teams whose win probability has fallen from relatively high levels to a low level.

This pattern of behavior is consistent with what prospect theory posits from actors who are making decisions from within the loss frame relative to their reference point. In particular, to the extent that teams that experience success early in the game have a higher reference point of success, these teams are more likely to frame their current situation as a loss from a prior reference point. Being within that loss frame would translate to a lower degree of loss aversion and thus a greater willingness to take risks. Also, consistent with the literature, our results show that the fourth down decisions of teams are generally too conservative relative to what would maximize win probability. As such, factors that increase a team's likelihood of attempting a fourth down play (such as being a heavier favorite or having experienced a high level of win probability earlier in the game) serve to improve the team's chances of winning through this channel of increased aggressiveness on fourth down.

In the context of fourth down decisions, coaches are more likely to go against the status quo of punting or attempting a field goal in situations where they were in a better position at a prior point in the game than where they currently find themselves. Coaches who are expected to win, and who experienced success earlier in the game, are more inclined to take the risky decision. In this context, these situations appear to encourage coaches to do what is actually optimal, meaning that success generates more success. On the other hand, coaches who were not expected to win the game, and who have not been at an advantageous point in the game tend to make suboptimal, risk-averse decisions (to punt or attempt a field goal rather than run a play); low expectations seem to encourage less risk-taking in situations where taking on risk is optimal. As such, lack of prior success induces suboptimal decision making. This is like the behavior of college football coaches in Owens and Roach (2018) who find that coaches who are in danger of being fired take on less risk even though it would be helpful from a win maximization perspective. Being in a weak position seems to introduce a psychological bias toward a suboptimal choice that seems less risky. This is also consistent with Paulsen (2021) who finds evidence that Major League Baseball players signed to long-term contracts shirk when they are in "party cities" on short rest days. These players are in an advantageous position and choose to take on excessively risky behaviors, whereas players with less job security engage in less risky behavior.

The presence of persistent inefficiencies in NFL fourth down decision making as well as the clear influence of behavioral factors on attitudes toward risk show that efficiently incorporating situational information is difficult even in situations where costs and benefits are relatively easy to quantify and where decisions are the subject to the scrutiny of many interested parties. Given that, these issues, and the behavioral phenomena that lead to them, likely extend to other strategic business settings where risky decisions and their outcomes are less clearly observable.

Notes

1. Generally speaking in a tournament setting, the magnitude of this discrete marginal benefit increases as the rank order improves. That is, the payoff structure has a convex shape.

2. This bias toward conservative behavior has been shown in other sports contexts, such as Paserman (2023), which shows professional tennis players who consistently hit safer shots that are higher over the net and closer to the center of the court on important points.

3. Mutual funds have an incentive to increase risk when their performance is just below the level that would be associated with a steep increase in payoffs. In this context, such funds include low performing funds (who would avoid relatively large outflows from their funds if they could bring their performance closer to the median fund) and high-performing funds (who would see very large inflows into their funds if they were to place among the very top performing funds in a given year).

4. The precise formula and details of the calculations are described in Pro Football Reference (n.d.) and can be found at: https://www.pro-football-reference.com/about/win_prob.htm. The win probability calculator itself can be found here: https://www.pro-football-reference.com/boxscores/win_prob.cgi.

5. This includes games hosted outside the United States, excluding a series of games the Buffalo Bills elected to play in Toronto (in those games the Buffalo Bills are designated as the home team).

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6. Note that 17.2% of fourth downs in primary subset involve a win probability below 10%, and 8% of the fourth downs in this subset involve a win probability above 90%.

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